

## QUESTION – 1

```
<!DOCTYPE html>
<html>
<head>
<title>Filter Palindrome</title>
</head>
<body>

<script>

function filterPalindrome(wordArray) {

let palindromeArray = [];

for (let i = 0; i < wordArray.length; i++) {

let currentWord = wordArray[i];
let reversedWord = currentWord.split('').reverse().join('');

if (currentWord === reversedWord) {
palindromeArray.push(currentWord);
}

}

return palindromeArray;
}

let words = ["madam", "racecar", "apple", "level", "hello"];

console.log(filterPalindrome(words));

</script>

</body>
</html>
```

## QUESTION – 2

```
<!DOCTYPE html>
<html>
<head>
<title>Second Largest</title>
</head>
<body>

<script>

function secondLargest(numberArray) {

let largestNumber = numberArray[0];
```

```

let secondLargestNumber = numberArray[0];

for (let i = 1; i < numberArray.length; i++) {

  if (numberArray[i] > largestNumber) {
    secondLargestNumber = largestNumber;
    largestNumber = numberArray[i];
  }
  else if (numberArray[i] > secondLargestNumber && numberArray[i] !== largestNumber) {
    secondLargestNumber = numberArray[i];
  }

}

return secondLargestNumber;
}

let numbers = [12, 45, 78, 23, 67];

console.log(secondLargest(numbers));

</script>

</body>
</html>

```

### QUESTION – 3

```

<!DOCTYPE html>
<html>
<head>
<title>Odd Numbers</title>
</head>
<body>

<script>

function generateOddNumbers(totalNumbers) {

  let oddNumberArray = [];
  let currentOdd = 1;

  for (let i = 0; i < totalNumbers; i++) {

    oddNumberArray.push(currentOdd);
    currentOdd = currentOdd + 2;

  }

  return oddNumberArray;
}

```

```
console.log(generateOddNumbers(5));
```

```
</script>
```

```
</body>
```

```
</html>
```

#### QUESTION – 4

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Count Vowels</title>
```

```
</head>
```

```
<body>
```

```
<script>
```

```
function countVowels(text) {
```

```
let vowelCount = 0;
```

```
let lowerCaseText = text.toLowerCase();
```

```
for (let i = 0; i < lowerCaseText.length; i++) {
```

```
let currentLetter = lowerCaseText[i];
```

```
if (
```

```
currentLetter == "a" ||
```

```
currentLetter == "e" ||
```

```
currentLetter == "i" ||
```

```
currentLetter == "o" ||
```

```
currentLetter == "u"
```

```
) {
```

```
vowelCount++;
```

```
}
```

```
}
```

```
return vowelCount;
```

```
}
```

```
console.log(countVowels("Hello World"));
```

```
</script>
```

```
</body>
```

```
</html>
```

## QUESTION – 5

```
<!DOCTYPE html>
<html>
<head>
<title>Reverse Words</title>
</head>
<body>

<script>

function reverseWords(sentence) {

let wordArray = sentence.split(" ");

let reversedSentence = wordArray.reverse().join(" ");

return reversedSentence;
}

console.log(reverseWords("Hello World"));

</script>

</body>
</html>
```

## QUESTION – 6

```
<!DOCTYPE html>
<html>
<head>
<title>Random Hex Color</title>
</head>
<body>

<button onclick="changeBackgroundColor()">Change Background</button>

<script>

function generateRandomHexColor() {

let colorLetters = "0123456789ABCDEF";

let hexColor = "#";

for (let i = 0; i < 6; i++) {

let randomIndex = Math.floor(Math.random() * 16);
```

```
hexColor = hexColor + colorLetters[randomIndex];

}

return hexColor;
}

function changeBackgroundColor() {

let randomColor = generateRandomHexColor();

document.body.style.backgroundColor = randomColor;

}

</script>

</body>
</html>
```